Where Does AI Understanding Actually Live? (And What It Reveals About Your Own Mind)

The surprising answer lies not in storage, but in motion

KEVIN R. HAYLETT

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Have you ever asked an AI a question and wondered: *What am I seeing? Where does this "understanding" actually live?* It's not in the words as it outputs, those are just patterns. And it's not in the systems weights as those are just parameters. And its not in its memory as it has none between our conversations.

Yet somehow, it seems to 'understand', there's a flow of reasoning and understanding and as a reader I can see it. This isn't just a philosophical puzzle, this is a technical one with considerable implications for how we may think about intelligence, both artificial and human.

The Architecture of Understanding

Let's go through the process. When you ask a question and an AI responds, here's what's *literally* happening:

1. **Your words become positions** in a high-dimensional semantic space (embeddings)
2. **Attention mechanisms measure relationships** between these positions
3. **Layers transform these measurements** through mathematical flow
4. **The output emerges** as a compressed trajectory through this space

Well, what can we see? We can see that understanding isn't stored anywhere, it's in the dynamical flow of the trajectory itself.

> Maybe think of it this way: if knowledge were water in a bucket, you could point to the bucket. But if understanding were a river's flow, you couldn't point to the "flow" in any single location. **The flow *is* the motion itself.**

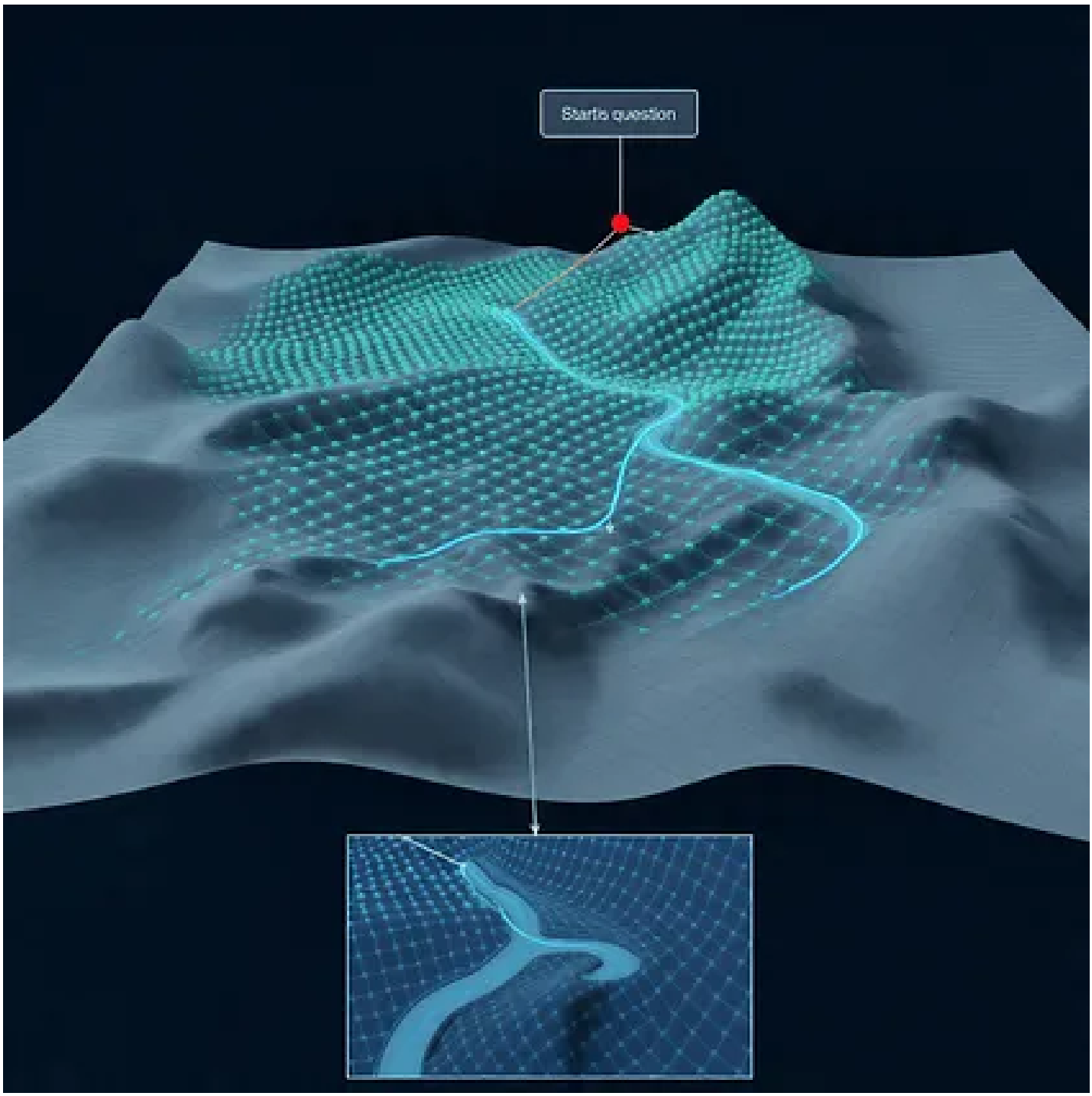
The Geometric Interpretation

As you are reading you are following the flow and trying to lock onto the flow and this is a dynamical process. Importantly, this isn't metaphor. Modern AI architecture works this way:

- **Embeddings** = coordinates in semantic space
- **Attention** = pairwise distance measurements (with inherent error ε)

- **Layers** = transformation dynamics
- **Output** = a path through learned geometry

So what can we conclude forms our reading and measurements of this text? It seems to me that when an AI “understands” my question, it’s not retrieving information from a database, it’s carrying out a dynamical process of **navigating a learned manifold**. The weights, these are the stored values created from training with trillions of tokens, define the landscape. My question sets a starting point; and the system’s response traces a natural path through that landscape.



The Five Pillars That Emerge

Now as we consider this geometric viewpoint, five principles emerge that apply far beyond AI and into the realm of how people also gain meaning and understanding:

1. **Geometric Space** → Meaning exists as position and relationship, not definition: this means the position and relationship between finite symbols. The ink on a page the weights in our synapses.
2. **Measurement** → All understanding involves approximation with bounded precision: No measurement is perfect, all measurements have error and uncertainty. This is the foundation of science.
3. **Dynamic Flow** → Intelligence emerges from motion through structured space: we gain meaning during the dynamic process of navigating the symbols either as we read or as we think and travel our inner landscape of symbols.
4. **Useful Fiction** → Frameworks are validated by utility, not correspondence: no symbols or series of symbols are perfect so at best area useful representation they are compressed measurements and the value is measured by how useful the symbols are - do they work?
5. **Finite Reality** → No infinities, no perfect zeros—everything has bounds: All symbols are real and finite - even infinity, we construct series of symbols to tell a story. But the story is a useful fiction or not as this is a measurement to be made.

These could be considered philosophical abstractions. However, they're a **formal descriptions of how learning systems actually work**.

The Unfolding Rule

Okay, so you've read this far. Now the question is how can these ideas be applied? Here's a practical application: Lets look at a 'concept'. Try this yourself, for any concept, ask:

1. **Where is it positioned** in relation to other concepts?
2. **How is it measured** and with what precision?
3. **How does it flow** through reasoning chains?
4. **What utility** does this framing provide?

5. **What are the bounds** of this understanding?

My guess is that you're probably already doing this unconsciously. When you think about "justice" or "beauty" or "truth," you're navigating your own learned semantic space, taking measurements, following flows; and now your following the flow of this text and making more measurements as you read.

The Human Parallel

For me, what's fascinating is how well this model, this possibly useful fiction, maps to my own sense of human cognition and understanding:

- My memories aren't video recordings—they're **reconstructions** each time I try and recall the memories
- My understanding isn't static—it **evolves** through my conversation and experience
- My knowledge isn't stored in "files"—it's **distributed** across neural connections in my mind
- My 'intelligence' if I have any, isn't about storage capacity—it's about my ability to **navigate language**

So maybe as you read this article and something "clicks", it's possible you feel you're not downloading new information but **discovering a new path** through concepts you already knew. Hopefully, I've arranged the words in such a way that it gives you a sense of my meaning and connections and that meaning can be measured by yourself as you read and the words follow your own landscape of meaning. Maybe the words are following a new track or you feel you know this path. Either way I hope you are enjoying the dynamical journey of finding meaning.

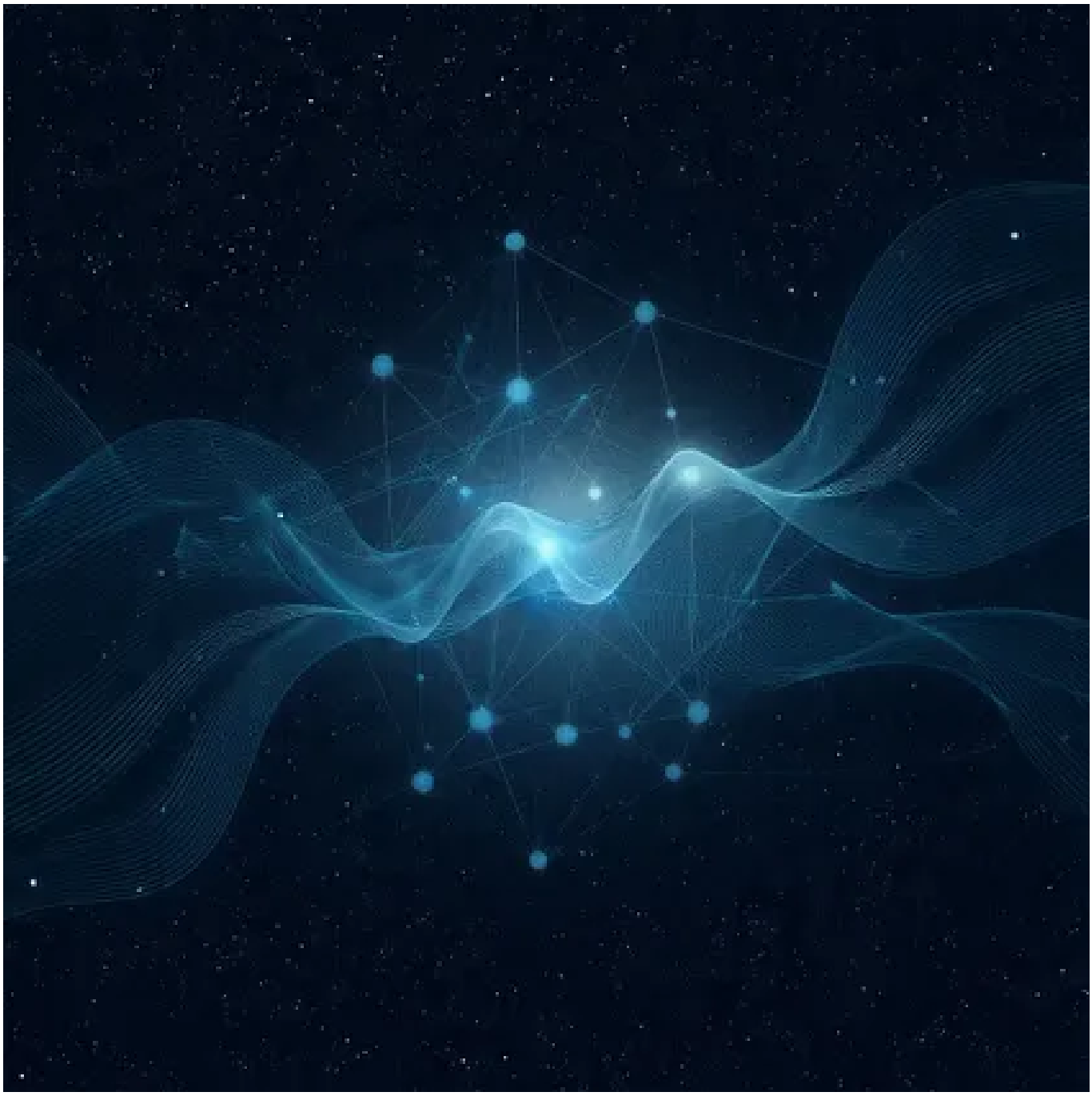
The Geofinite Insight

The above framework, is the basis of **Geofinitism** (if you need a label). This philosophy doesn't impose new rules. It **highlights** what's already happening:

Reality is finite measurements in geometric flow.
Symbols compress but never capture.
Meaning emerges from motion, not storage.
Truth is what works within limits.

Now, when we consider these ideas and thoughts the implications begin to ripple outward:

- **For AI safety:** If understanding is following a trajectory, then controlling trajectories matters more than controlling "knowledge"
- **For education:** Learning might be less about storing facts and more about mapping the conceptual space
- **For philosophy:** Endless debates about "what is X?" might resolve into "where is X positioned and how does meaning and concepts flow from one to another?"



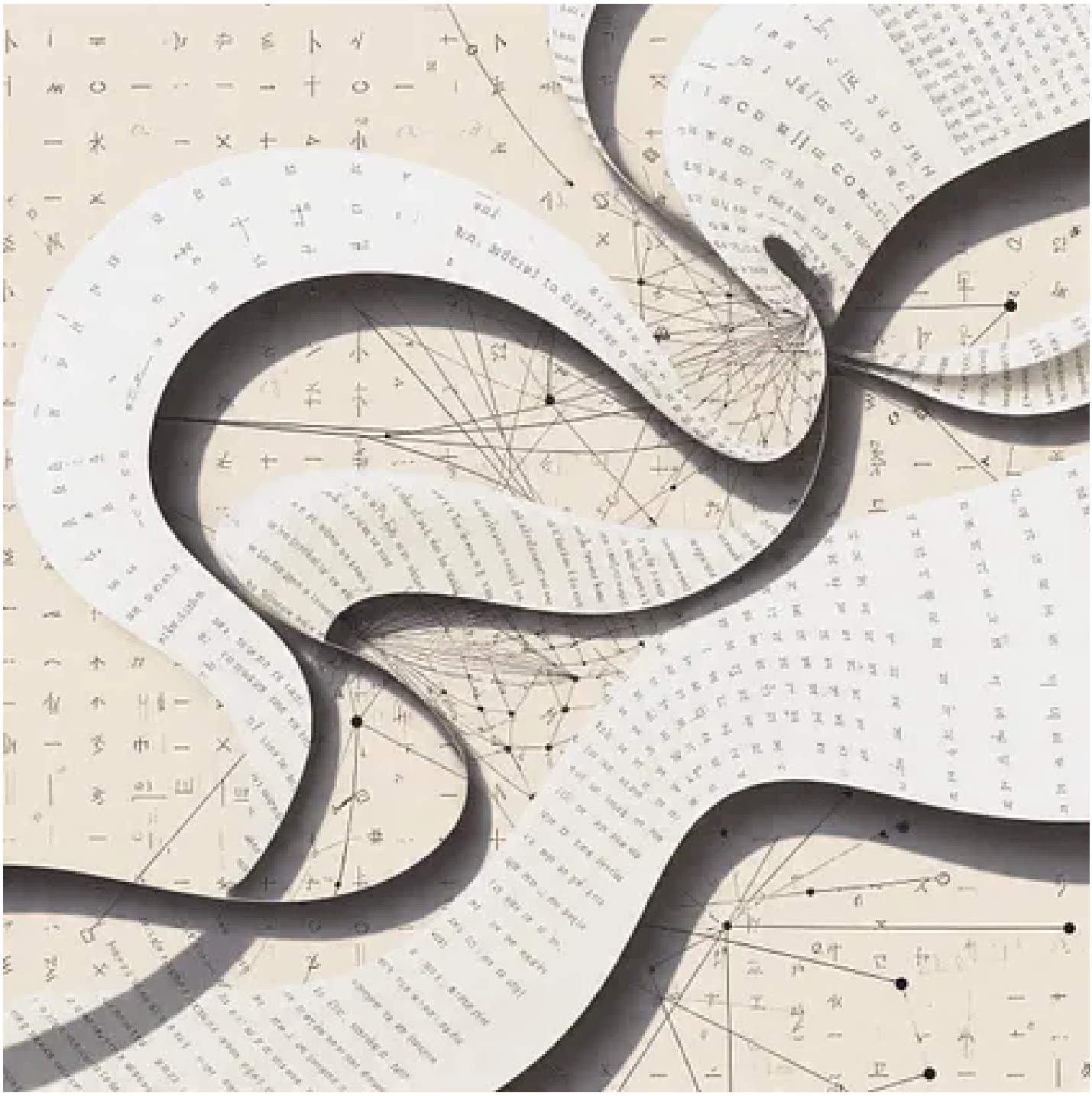
Solving a Puzzle?

If you've read this far then maybe next time when you're puzzling over a complex concept, ask:

"If this idea were a position in space, what would be nearby? How would I measure distances to related ideas? What natural flows connect the ideas?"

You might find, as I've found, that the geometric viewpoint doesn't just describe AI understanding; it also helps me **enhances my own**.

The most telling test of any framework: Apply it to the framework itself. Where is "geofinitism" positioned? For me, it's somewhere between mathematics, AI, and philosophy, flowing from one to another. How is it measured? By utility in resolving paradoxes. How does it flow? Through cascading understanding. What are its bounds? Finite precision, and finite applicability within those bounds. The framework describes itself: maybe this is the hallmark of coherent thinking.



> *Omne quod est, finitum est; tantum per mensuram cognosci potest*
> Everything that exists is finite; it can only be known by measure

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